Section 3.8 — Rational Exponents

Math 103: College Algebra

As reference when completing this assignment, recall that we define

$$x^{p/q} = \sqrt[q]{x^p} = \left(\sqrt[q]{x}\right)^p$$

1 Translating Radicals to Rational Exponents

Rewrite each expression using rational exponents. Use negative exponents rather than writing variable expressions in denominators. Assume all variables represent positive real numbers.

1. $\sqrt[4]{25}$
2. $\sqrt[7]{r^5}$
3. $\dfrac{1}{\sqrt[7]{6^2}}$
4. $\dfrac{8}{\sqrt[7]{n^9}}$
5. $\left(\sqrt[3]{5}\right)^7$
6. $\left(\sqrt[3]{m}\right)^8$
7. $\sqrt[7]{(5w+3)^5}$
8. $\sqrt[4]{(2x^2-1)^3}$

2 Translating Rational Exponents to Radicals

Rewrite each expression in radical notation. (For number bases, don't evaluate, just rewrite.) Your answer should include only positive integer exponents.

9. $27^{1/3}$
10. $y^{1/4}$
11. $-x^{3/4}$
12. $8n^{2/3}$
13. $x^{5/2}$
14. $3y^{-2/3}$
15. $(4x)^{-1/2}$
16. $(xy)^{-8/5}$

3 Computing Rational Powers

Compute each power by hand. You can use the following reference:

x	x^2	x^3	x^4	x^5	x^6
1	1	1	1	1	1
2	4	8	16	32	64
3	9	27	81	243	729
4	16	64	256	1,024	4,096
5	25	125	625	3,125	15,625
6	36	216	1,296	7,776	46,656

If the given power is not a real number, write "undefined." All answers for this section are rational.

17. $64^{1/6}$
18. $\left(\dfrac{1}{8}\right)^{-2/3}$
19. $(-32)^{-3/5}$
20. $(-64)^{3/2}$
21. $\left(\dfrac{8}{27}\right)^{-1/3}$
22. $\left(-\dfrac{25}{49}\right)^{5/2}$
23. $\left(\dfrac{81}{625}\right)^{3/4}$
24. $\left(-\dfrac{125}{216}\right)^{4/3}$

Write each expression in simplified radical form, including rationalizing any denominators.

25. $2^{2/3}$
26. $4^{5/3}$
27. $3^{-3/2}$
28. $108^{-1/2}$

4 Simplifying with Rational Exponents

Simplify each expression as much as possible. Your answer should include a single occurrence of each variable and only positive exponents. Leave your answer in terms of rational exponents. To avoid concerns about where expressions are defined, assume all variables represent positive real numbers.

29. $\left(xyz^2\right)^{1/3}$
30. $\left(x^{2/3}\right)^{3/8}$
31. $x^{2/3}x^{1/6}x^{1/2}$
32. $\left(x^{7/6}y^{-5/8}\right)^{-12/5}$
33. $\dfrac{x^{1/2}y^{1/3}}{x^{2/3}y^{1/4}}$
34. $\dfrac{x^{-1/3}y^{2/3}z^{-1/6}}{x^{1/2}y^{1/2}z^{1/2}}$
35. $\left(\dfrac{x^{1/2}y^{1/5}}{xy^2}\right)^{4/3}$
36. $\left(\dfrac{x^{-1}y^{-1/2}z^{1/8}}{x^{-4/3}y^{-1/8}z^{3/4}}\right)^{-4/7}$